SUPPLEMENTARY MATERIAL

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1. PROOF TO THEOREM 1

Theorem 1. *Let $(\tilde{X}, \bar{X}, Y)$ be random variables and let g be a (possibly stochastic) encoder applied to both $\tilde{X}$ and $\bar{X}$. Assume the encoder is sufficient [1], then:*

$$I\big(g(\tilde{X}); Y \mid g(\bar{X})\big) = I\big(\tilde{X}; Y \mid \bar{X}\big). \tag{1}$$

Proof. Let $(\tilde{X}, \bar{X}, Y)$ be random variables and let g be a (possibly stochastic) *shared* encoder applied argument-wise to $\tilde{X}$ and $\bar{X}$. Assume *sufficiency*:

$$P(Y \mid \bar{X}) = P\big(Y \mid g(\bar{X})\big), \tag{S1}$$

$$P(Y \mid \tilde{X}, \bar{X}) = P\big(Y \mid g(\tilde{X}), g(\bar{X})\big). \tag{S2}$$

By the chain rule of conditional mutual information,

$$I(\tilde{X}; Y \mid \bar{X}) \;=\; H(Y \mid \bar{X}) - H(Y \mid \tilde{X}, \bar{X}). \tag{2}$$

By sufficiency (S1) and (S2),

$$H(Y \mid \bar{X}) = H\big(Y \mid g(\bar{X})\big), \tag{3}$$

$$H(Y \mid \tilde{X}, \bar{X}) = H\big(Y \mid g(\tilde{X}), g(\bar{X})\big). \tag{4}$$

Therefore,

$$I(\tilde{X}; Y \mid \bar{X}) = H\big(Y \mid g(\bar{X})\big) - H\big(Y \mid g(\tilde{X}), g(\bar{X})\big) \tag{5}$$

$$= I\big(g(\tilde{X}); Y \mid g(\bar{X})\big), \tag{6}$$

which proves the claim.

□

2. INTUITIVE EXPLANATIONS TO KEY EQUATIONS

Intuitively, Eq. (7) characterizes the causal relevance of a selected feature subset $\tilde{x}$ by measuring the additional predictive information it provides about the model output y, conditioned on the remaining features $\bar{x}$. If $I(\tilde{x}; y \mid \bar{x})$ is large, the prediction cannot be well recovered without $\tilde{x}$, indicating a strong causal contribution.

Based on this notion, Eq. (8) augments the prediction-fidelity objective with a CMI-based regularization term. This term discourages the selection of features that are redundant given other inputs and favors subsets that uniquely contribute to the prediction.

Direct estimation of CMI in the input space is generally infeasible for high-dimensional or structured data. Therefore, Eq. (9) estimates CMI in the latent space of the black-box model. Under the encoder sufficiency assumption, this approximation preserves the causal interpretation while enabling efficient optimization.

Finally, Eq. (12) shows that CMI can be interpreted as a discrepancy between two conditional output distributions, with and without access to $\tilde{x}$, highlighting its role in quantifying the predictive impact of the selected features.

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3. ADDITIONAL RESULTS

3.1. Comparison with Post-hoc Explanation Methods

To further evaluate the quality of the proposed explanations, we compare our instance-wise feature selection approach with representative post-hoc explanation methods, including LIME [2] and SHAP [3], on a multimodal image dataset constructed by pairing MNIST and SVHN images of the same digit class. We focus on MNIST since fidelity and consistency can be naturally and reliably defined for image-based inputs, whereas such metrics are less standardized for multi-omics tabular data.

We first train a multimodal classifier (Multimodal Information Bottleneck, MIB) and treat it as a fixed black-box model to be explained. Our explainer is then trained post-hoc to select instance-wise important features by minimizing the divergence between the black-box predictions on the original input and the masked input. For a fair comparison, LIME and SHAP are also applied post-hoc to the same fixed black-box classifier. Since LIME/SHAP are designed for single-input explanations, we apply them separately to each modality and report the modality-wise explanation quality using the same evaluation protocol.

We evaluate explanation quality using two commonly adopted criteria:

- **Fidelity (average confidence drop).** Fidelity is measured as the average decrease in the model's prediction confidence when only the features selected by the explainer are retained as input (i.e., all other features are masked). A lower confidence drop indicates that the selected features better preserve the black-box decision.

- **Consistency (maximum sensitivity).** Consistency measures the maximum change in the explanation under small perturbations of the input. Lower sensitivity indicates more stable and robust explanations.

Table 1 reports the quantitative comparison on MNIST. Our method achieves lower confidence drop (higher fidelity) and lower sensitivity (better consistency) than LIME and SHAP under the same black-box classifier and evaluation protocol, indicating more faithful and stable instance-wise explanations.

Table 1: Comparison between ours and post-hoc explanation methods on a multimodal image dataset constructed by pairing MNIST and SVHN images of the same digit class. Lower values indicate better performance.

Method	Fidelity (↓)	Consistency (↓)
LIME	9.1	0.32
SHAP	10.8	0.35
Ours	**7.0**	**0.24**

3.2. Evaluation on combining sMRI and fMRI for Major Depressive Disorder Diagnosis

We further evaluate our method on the REST-meta-MDD dataset [4], which consists of structural MRI (sMRI) and resting-state fMRI (rs-fMRI) data for binary classification between major depressive disorder patients and healthy controls. Following standard preprocessing pipelines, sMRI data are represented as voxel-based gray matter volume features, while rs-fMRI data are modeled as brain connectivity graphs.

A multimodal Information Bottleneck (IB) classifier [5] is first pretrained using both modalities and treated as a fixed black-box model. Our post-hoc explainer is then applied to generate instance-wise feature selections for both sMRI and fMRI.

Beyond predictive performance, we focus on the stability of the resulting explanations across independent runs. Figure 1 reports between-run agreement measured by the Jaccard similarity of selected feature sets, defined as the overlap between feature subsets selected in two runs, $J(S_i, S_j) = |S_i \cap S_j|/|S_i \cup S_j|$, as well as group-level recurrence of frequently selected features. We observe that fMRI-based explanations exhibit higher stability than sMRI-based ones, while both modalities demonstrate non-random group-level consensus. This difference is expected given the structured nature of functional connectivity graphs compared to volumetric feature representations.

The black-box multimodal classifier achieves a classification accuracy of 69.8 ± 1.92. When predictions are made using only the features selected by the explainer from a single modality, the accuracy is 65.1 ± 1.01 for fMRI and 67.3 ± 1.78 for sMRI. As a control experiment, when the same number of features are randomly removed from fMRI and sMRI, the classification accuracy drops substantially to 53.9 ± 2.33 and 55.6 ± 1.12, respectively. These results indicate that the features selected by the explainer preserve substantially more predictive information than random feature removal, while each single modality alone remains less informative than the full multimodal model.

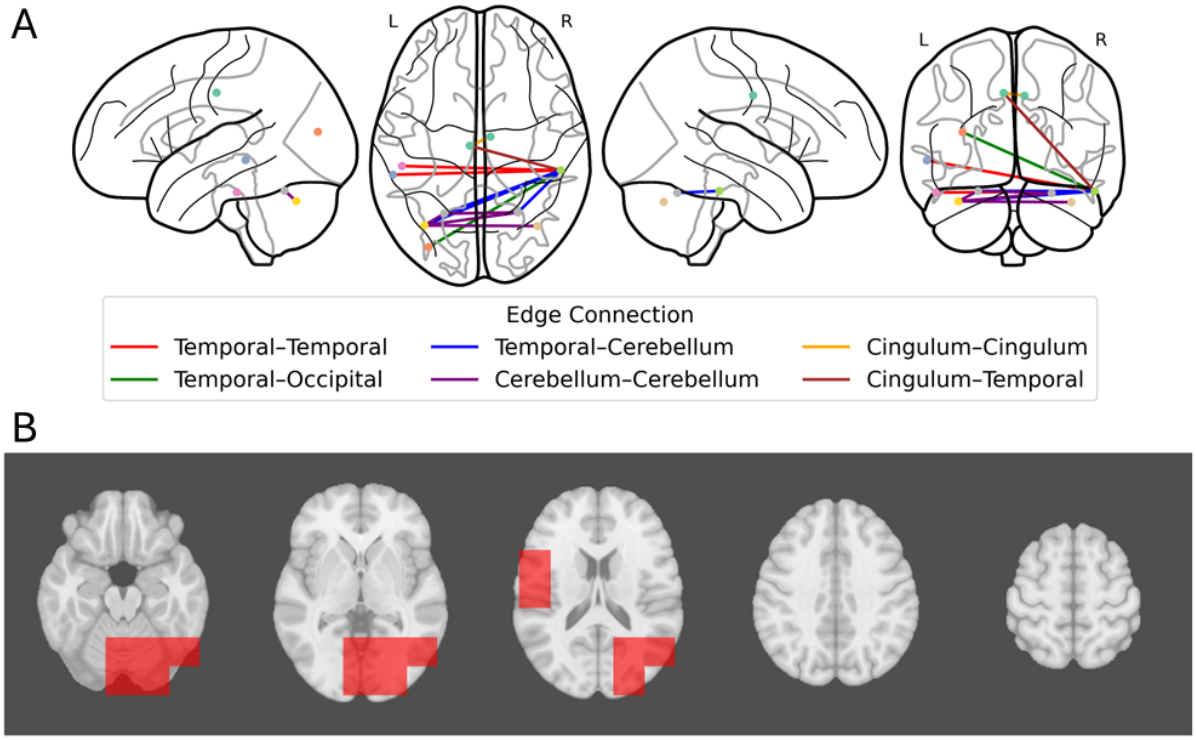

Fig. 1: Consensus feature selections across runs and subjects. (A) fMRI consensus edges rendered on the cortical surface, where edge colors indicate edge categories and node colors are used for ROI identification across different axial views. (B) sMRI consensus subvolumes (highlighted in red) overlaid on axial slices of the MNI template. Consensus is defined as features consistently selected across all runs within each diagnostic group.

Overall, these results demonstrate that the proposed method yields stable and reproducible explanations on complex, structured multimodal neuroimaging data.

3.2.1. Experimental details

We provide detailed descriptions of the dataset, preprocessing pipelines, model architectures, and training procedures for the multimodal neuroimaging experiments.

3.2.1.1. Dataset and Preprocessing

We use the REST-meta-MDD dataset, a large-scale multi-site neuroimaging dataset including structural MRI (sMRI) and resting-state fMRI (rs-fMRI) data collected from 25 cohorts (17 hospitals), totaling 2,428 participants (1,300 major depressive disorder (MDD) patients and 1,128 healthy controls (HC)). Diagnoses followed DSM-IV/ICD-10 criteria, and MDD and HC groups were matched on age and sex.

For rs-fMRI data, consortium preprocessing pipelines (DPARSF/SPM12) were applied, including discarding the first 10 volumes, slice-timing correction, motion correction, normalization to MNI space, temporal band-pass filtering (0.01-0.10 Hz), and nuisance regression to remove confounding signals (head motion, global brain signal, white matter, cerebrospinal fluid, and linear and quadratic drifts). Additional quality control procedures were applied to exclude subjects with incomplete time series, poor spatial normalization, excessive head motion, or cohorts with fewer than ten subjects per group, resulting in 1,604 usable rs-fMRI subjects (848 MDD, 794 HC).

For sMRI data, all 2,428 participants were retained. A VBM-style preprocessing pipeline was used to generate gray-matter volume (GMV) maps, including bias-field correction, GM/WM/CSF segmentation, MNI normalization, and cross-site intensity normalization. The resulting GMV images ($121 \times 145 \times 121$ voxels) were z-normalized per subject.

3.2.1.2. Functional Connectivity Graph Construction

We represent rs-fMRI data as graphs for graph-based learning. REST-meta-MDD provides ROI-averaged BOLD time series parcellated using the AAL-116 atlas. For each subject, we compute a 116×116 Pearson correlation matrix and apply Fisher's z-transformation to improve the distributional properties of the connectivity values. Each entry represents the temporal coupling strength between a pair of ROIs.

To define graph edges, we average subject-level functional connectivity matrices into a dataset-level matrix and retain the top 20% of correlations, reducing 13,456 possible ROI pairs to 2,660 edges. Only positive correlations are retained, as negative correlations may reflect artifacts introduced by global signal regression. Thresholding at the dataset level yields a shared binary adjacency matrix across subjects, enabling consistent downstream analysis of selected edges. The final graph representation consists of 116 nodes with node features given by full ROI correlation profiles.

3.2.1.3. Model Architecture

We employ a multimodal Information Bottleneck (IB) framework to jointly encode sMRI and rs-fMRI data. Separate modality-specific encoders are used to extract latent representations, which are then optimized under the IB objective and treated as a fixed black-box model for post-hoc explanation.

The rs-fMRI connectivity graph is encoded using a three-layer Graph Convolutional Network (GCN) composed of GCNConv, ReLU activation, and dropout ($p = 0.15$). The graph encoder maps node features from dimensions $116 \to 116 \to 96 \to 64$. The resulting 64-dimensional node embeddings are ℓ_2-normalized and aggregated using second-order pooling (SOPOOL). A three-layer MLP ($64 \to 32 \to 32 \to 32$ with ReLU) is applied, followed by a bilinear outer product $H^\top H \in \mathbb{R}^{32\times 32}$, which is flattened to obtain a 1024-dimensional graph-level representation.

The sMRI encoder follows a similar overall design but is adapted to three-dimensional volumetric inputs. We adopt a five-layer 3D CNN architecture inspired by SFCN, using kernel size 3, stride 1, and padding 1. The first four convolutional layers are followed by BN–ReLU and $2 \times 2 \times 2$ max-pooling, progressively reducing spatial resolution: $121 \times 145 \times 121 \to 60 \times 72 \times 60 \to 30 \times 36 \times 30 \to 15 \times 18 \times 15 \to 7 \times 9 \times 7$. The fifth convolution projects the channel dimension to 64 without pooling, yielding a feature map of size $[B, 64, 7, 9, 7]$. Global average pooling is applied, followed by a projection head with dropout ($p = 0.5$) and two fully connected layers with GELU activation ($64 \to 64 \to 64$) to obtain a 64-dimensional subject embedding.

3.2.1.4. Training Details

All models are trained for 300 epochs with a batch size of 64, including a 65-epoch warm-up phase. The Adam optimizer is used for both the graph encoder (learning rate 10^{-3}, weight decay 10^{-4}) and the feature selector (learning rate 10^{-2}, weight decay 10^{-6}), together with cosine learning rate annealing. The Information Bottleneck regularization weight is set to $\lambda_{\mathrm{MI}} = 0.08$.

3.3. Sensitivity Analysis on Hyperparameters

In this section, we analyze the sensitivity of our post-hoc explanation framework to two key hyperparameters on the ROSMAP dataset: (i) the feature subset size k, which controls the sparsity of the instance-wise explanations, and (ii) the causal regularization coefficient β in Eq. (9), which controls the strength of the instance-wise causal constraint.

Since our explainer is applied post-hoc to a fixed black-box classifier, we evaluate accuracy *without retraining* the classifier. Specifically, for each test instance, we retain only the features selected by the explainer in each modality and mask all remaining features, then feed the masked input into the original multimodal classifier. Classification accuracy is computed by comparing the resulting predictions with the ground-truth labels. All results are reported as mean $\pm$ standard deviation over 10 independent runs. Under the default hyperparameter setting used throughout the paper ($k = 30$ features per modality and $\beta = 0.1$), the masked-input accuracy on ROSMAP is 82.5 ± 0.9.

3.3.1. Sensitivity to Feature Subset Size k

We analyze the effect of the feature subset size k by varying the number of selected features per modality while fixing $\beta = 0.1$. As reported in Table 2, when k is too small, classification accuracy drops noticeably, since insufficient features are retained to preserve the predictive behavior of the black-box model. As k increases, accuracy improves and quickly reaches a stable plateau. Further increasing k beyond this range yields only marginal gains, as the additional selected features are largely redundant and do not provide extra generalizable information. This behavior reflects the expected trade-off between explanation sparsity and prediction fidelity.

Table 2: Sensitivity to feature subset size k on the ROSMAP dataset. Accuracy is evaluated on masked inputs using the fixed black-box classifier (mean $\pm$ std over 10 runs).

k (per modality)	Accuracy (%) $\uparrow$
10	74.2 ± 2.4
20	81.5 ± 2.3
30	$\mathbf{82.5 \pm 0.9}$
40	82.5 ± 1.0
60	82.4 ± 1.1

3.3.2. Sensitivity to the Causal Regularization Coefficient β

We further study the effect of the causal regularization coefficient β by fixing $k = 30$ and varying β over a range of values. Here, β explicitly controls the strength of the instance-wise causal constraint in Eq. (9), rather than serving as a generic regularization weight. As shown in Table 3, we observe a clear "sweet spot": when β is too small, the causal constraint is weak and the explainer may select redundant features; when β is too large, over-emphasizing the causal term can harm prediction fidelity. Overall, accuracy remains stable over a broad intermediate range of β, indicating robustness to the choice of causal regularization strength.

Table 3: Sensitivity to the causal regularization coefficient β on the ROSMAP dataset, with $k = 30$ fixed. Accuracy is evaluated on masked inputs using the fixed black-box classifier (mean $\pm$ std over 10 runs).

β	Accuracy (%) $\uparrow$
0	80.9 ± 1.8
0.01	81.8 ± 1.3
0.05	82.2 ± 1.1
0.10	$\mathbf{82.5 \pm 0.9}$
0.20	81.7 ± 1.4

4. REFERENCES

[1] Yonglong Tian, Chen Sun, Ben Poole, Dilip Krishnan, Cordelia Schmid, and Phillip Isola, "What makes for good views for contrastive learning?," *Advances in neural information processing systems*, vol. 33, pp. 6827–6839, 2020.

[2] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin, "" why should i trust you?" explaining the predictions of any classifier," in *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, 2016, pp. 1135–1144.

[3] Scott M. Lundberg and Su-In Lee, "A unified approach to interpreting model predictions," in *Proceedings of the 31st International Conference on Neural Information Processing Systems*, 2017, p. 4768–4777.

[4] Chao-Gan Yan, Xiao Chen, Le Li, Francisco Xavier Castellanos, Tong-Jian Bai, Qi-Jing Bo, Jun Cao, Guan-Mao Chen, Ning-Xuan Chen, Wei Chen, et al., "Reduced default mode network functional connectivity in patients with recurrent major depressive disorder," *Proceedings of the National Academy of Sciences*, vol. 116, no. 18, pp. 9078–9083, 2019.

[5] Sijie Mai, Ying Zeng, and Haifeng Hu, "Multimodal information bottleneck: Learning minimal sufficient unimodal and multimodal representations," *IEEE TMM*.